

SYNAPSE-SOC: LIVE STREAMING SOC ARCHITECTURE

7-Phase Engineering Verification & Operational Runbook | Microsoft Innovate 2026

1. Executive Summary & Architectural Transition

SYNAPSE-SOC has transitioned from an offline batch simulation tool into a **live event-driven, streaming SOC detection engine**. The platform continuously ingests telemetry across three live sources (Windows Security Event Logs, File Integrity Monitor with Watchdog, and SSH Deception Honeypot), normalizes them into a standardized **Common Alert Schema**, evaluates events against **5 stateful behavioral detection rules**, correlates multi-stage attacks over a **15-minute sliding window** using graph connected components, prioritizes incidents via **CADR Asset-Criticality metrics**, and renders real-time insights across a **Rich Live Terminal Console** and a **FastAPI Web Console** with interactive Telegram containment approval.

2. 7-Phase Engineering & Deliverables Matrix

Phase & Focus Area	Key Implementation Deliverables	Core Source Files	Status
Phase 1: Foundation Common Schema & Queue	Normalized schema (`source, type, host, user, src_ip, timestamp, severity`), `UnifiedEventQueue`, `BaseSource(ABC)`	`engine/models.py` `sources/base_source.py` `sources/event_queue.py`	[DONE]
Phase 2: Windows Event Log Local OS Telemetry	Event IDs 4625 (Failed), 4624 (Success), 4688 (Process Spawn), WinEvt XML/dict parser, Auditpol helper	`sources/windows_events.py` `sources/windows_auditing.py`	[DONE]
Phase 3: FIM Watchdog Ransomware Detection	Watchdog on `protected_assets/`, mass change (>20 files/10s), `.locked/.crypto` regex, test harness	`sources/fim_source.py` `test_fim_harness.py`	[DONE]
Phase 4: Streaming Brain Stateful Sliding Correlator	`on_new_alert(alert)` with 15-min sliding window, 5 core detection rules (T1110, T1078, T1059, T1486, T1046), CADR risk progression	`engine/streaming_correlator.py` `engine/detection_rules.py` `engine/risk_scorer.py`	[DONE]
Phase 5: Live Terminal UI Rich Live Console	Split-pane `rich.live.Live` console, top HUD bar (Events In, Incidents Out, Noise Compression %), explicit Tier Badges (T0-T3), 3-bullet AI brief	`live_dashboard.py` `engine/summarizer.py`	[DONE]
Phase 6: Notifications Mobile Escalation & HITL	Async Telegram/Discord bot, 10-min deduplication, grounded 3-bullet summary with cited Alert IDs, `[Block IP]` human confirmation	`notifications/telegram_bot.py` `notifications/notifier.py`	[DONE]
Phase 7: Honeypot & Polish Deception, IP & Replay	SSH deception honeypot (port 2222), Cowrie adapter, AbuseIPDB/GeoIP enrichment, `replay_mode.py`, 3-min end-to-end demo script	`sources/ssh_honeypot.py` `engine/ip_enrichment.py` `demo_attack_simulation.py`	[DONE]

3. Left Panel UI Badge Fix & CADR Risk Model

Left Panel Asset Tier Badge Fix: In the Web Console (`web/static/index.html`) and Terminal UI (`live_dashboard.py`), every incident row now clearly renders a color-coded asset tier badge:

- **T0 (Crown Jewel):** Red/Purple badge with 3.0x CADR Multiplier (Domain Controllers, Key Identity Vaults).
- **T1 (Mission Critical):** Amber badge with 2.0x CADR Multiplier (Core Databases, File Shares, Production APIs).
- **T2 (Internal Business):** Blue badge with 1.0x Multiplier (Corporate Workstations, Internal Services).
- **T3 (Dev / Lab / Honeypot):** Gray badge with 0.5x Multiplier (Isolated Sandboxes, Deception Traps).

The sidebar filter dropdown and search bar seamlessly match by both badge codes (`T0`) and full names.

4. 5 Core Stateful Detection Rules

Rule Name & MITRE Tag	Trigger Condition (15-min Sliding Window)	Severity	CADR Impact
1. Credential Brute Force T1110.001	≥ 5 failed logon attempts (Event 4625) within 5 mins for same user/IP	HIGH	+25 pts (Access)
2. Post-Brute Valid Logon T1078 Valid Accounts	Successful logon (Event 4624) within 15 mins of ≥ 3 failed logon attempts	CRITICAL	+35 pts (Privilege)
3. Spawn After Logon T1059 Command Interpreter	PowerShell / CMD / Mimikatz spawned within 5 mins of interactive logon	CRITICAL	+25 pts (Execution)
4. Ransomware Velocity T1486 Data Encrypted	FIM Watchdog detects > 20 file renames/mods in 10s or <code>.locked</code> extension	CRITICAL	+40 pts (Impact)
5. Honeypot Deception Probe T1046 Service Discovery	Direct inbound TCP/SSH connection attempt on isolated deception port 2222	HIGH	+20 pts (Recon)

5. Operator Runbook & Verification Commands

Execution Goal	Command Line (PowerShell / Command Prompt)
Interactive Launcher (All Modes)	<code>`. \run_demo.bat`</code>
3-Minute End-to-End Attack Demo	<code>`python demo_attack_simulation.py` (or <code>`python main.py --demo`</code>)</code>
Live Terminal Dashboard	<code>`python main.py --live`</code>
Safe FIM Ransomware Harness	<code>`python test_fim_harness.py`</code>
FastAPI Web SOC Console	<code>`python main.py --web` (Navigate to <code>http://localhost:8000</code>)</code>
Recorded Attack Replay Mode	<code>`python replay_mode.py --dataset data/captured_attacks.json --delay 0.5`</code>
Automated Test Suite (15 Unit Tests)	<code>`python -m unittest discover -s tests`</code>

6. Safety Protocol & Human-in-the-Loop Response

- **Human-in-the-Loop Safety Gate:** All critical mitigation playbooks (such as [Block IP] and [Isolate Host]) strictly require manual analyst authorization before perimeter firewall ACL updates or EDR network isolation commands are dispatched.
- **Safe Test Isolation:** The FIM simulation harness operates exclusively within isolated mock directories (`protected_assets/`) and performs automated cleanup.
- **Credential Security:** Zero sensitive API tokens or secrets are hardcoded; all integrations rely on environment variables.